ECO 202 — In-Class Exam 3

Worked solutions

Fall 2026 | Classes 10–13

Equivalent exact expressions and valid alternative reasoning are acceptable. For practice study, attempt the paper first, then compare reasoning, close these solutions, and reconstruct any missed method independently.

Questions 1–4: multiple choice

Question	Answer	Reason and distractor check
1	B	The interval has width 1 and density 1/2, hence probability 1/2. A is false because a continuous point has probability zero. C is false because $\mathbb{P}(T \leq 1) = 1/2$. D confuses density with mass.
2	B	The sample SD is $s = \sqrt{36} = 6$, so the estimated SE is $s/\sqrt{25} = 6/5$. A gives observation spread; C gives the estimated variance of the sample mean; D divides the SD by n instead of $\sqrt{n}$.
3	A	$\mathrm{SE}(\bar{X}_n) = \sigma/\sqrt{n}$, so the ratio is $\sqrt{25/100} = 1/2$. B confuses variance and SE; C reverses the direction; D ignores averaging.
4	D	Unbiasedness means centering at the target over repetitions. A confuses a realized error with bias. B ignores variance and the MSE tradeoff. C asserts an external-population claim not supplied by the model.

5. Joint outcomes and combined risk

(a) The column sums are $\mathbb{P}(X = 0) = 1/2$ and $\mathbb{P}(X = 2) = 1/2$. Thus

$$\mathbb{E}[X] = 0(1/2) + 2(1/2) = 1, \qquad \mathbb{E}[X^2] = 2, \qquad \mathrm{Var}(X) = 2 - 1^2 = 1.$$

(b) Only the $X = 2$ column contributes to the product expectation:

$$\mathbb{E}[XY] = (-2)(1/5) + (2)(3/10) = 1/5.$$

Therefore

$$\mathrm{Cov}(X, Y) = \frac{1}{5} - (1)\frac{3}{5} = -\frac{2}{5}, \qquad \mathrm{Corr}(X, Y) = \frac{-2/5}{(1)(4/5)} = -\frac{1}{2}.$$

The negative sign records opposite movement of deviations from the respective means. It is not a causal conclusion.

(c) The covariance term remains because these variables are dependent and have nonzero covariance:

$$\mathrm{Var}(X + Y) = 1 + \frac{16}{25} + 2\left(-\frac{2}{5}\right) = \frac{21}{25}.$$

Omitting covariance would give 41/25, overstating the variance because some fluctuations offset one another. Directly, $X + Y$ equals 1 with probability 7/10 and 3 with probability 3/10, which verifies the same answer.

6. One sampling distribution, several criteria

(a) The four equally likely samples are

Sample	Median
$\{1,3,4\}$	3
$\{1,3,5\}$	3
$\{1,4,5\}$	4
$\{3,4,5\}$	4

Thus $\mathbb{P}(T=3)=\mathbb{P}(T=4)=1/2$.

(b) The target is $\mu=(1+3+4+5)/4=13/4$. Meanwhile $\mathbb{E}[T]=(3+4)/2=7/2$. Its bias for the *mean* is $7/2-13/4=1/4$. The target must remain visible: the question does not ask whether T estimates a population median.

(c) The two possible medians are each $1/2$ from their expectation, so $\mathrm{Var}(T)=1/4$. The decomposition gives

$$\mathrm{MSE}(T;\mu)=\frac{1}{4}+\left(\frac{1}{4}\right)^2=\frac{5}{16}.$$

Equivalently, the realized errors are $-1/4$ and $3/4$, each with probability $1/2$, whose squared average is $5/16$.

(d) First, σ^2/n is the independent *sample-mean* variance formula, whereas T is a median. Second, the sample is drawn *without replacement*: distinct draw values are dependent, so even the sample mean's variance needs a design-specific calculation. Enumerating the actual mechanism avoids both mistakes; no finite-population-correction formula is needed.

7. Counts, proportions, and approximation

(a) Because $H_i^2=H_i$,

$$\mathbb{E}[H_i]=\frac{1}{2},\qquad \mathrm{Var}(H_i)=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}.$$

Linearity gives $\mathbb{E}[K]=100/2=50$, without requiring independence for this expectation. Independence makes the cross-covariances zero, giving $\mathrm{Var}(K)=100/4=25$. Consequently $\mathbb{E}[\widehat{P}]=1/2$ and

$$\mathrm{Var}(\widehat{P})=\frac{25}{100^2}=\frac{1}{400},\qquad \mathrm{SE}(\widehat{P})=\frac{1}{20}.$$

(b) The integer event $K\geq 60$ becomes the continuous event above 59.5. Its standardized boundary is

$$z=\frac{59.5-50}{5}=1.90,\qquad \mathbb{P}(K\geq 60)\approx 1-\Phi(1.90)=0.0287.$$

The exact count is binomial; the continuous Normal shape is an approximation. Expected successes and failures are both 50, supporting use of that approximation here. The continuity correction aligns the integer boundary but does not make the distributions identical.

(c) Under continued independent sampling, for every fixed $\varepsilon > 0$,

$$\mathbb{P}(|\widehat{P}_n - 1/2| > \varepsilon) \longrightarrow 0.$$

No. If $H_{101} = 1$, then $\widehat{P}_{101} = 61/101 > 0.6$, so the next realized proportion moves farther from $1/2$. The theorem is a repeated-sampling concentration statement, not a monotone-path guarantee. A qualitative explanation suffices; this numerical counterexample is not required.

(d) Now $\widehat{P} = (100H)/100 = H$. Its variance is $1/4$ and its standard error is $1/2$. Recording copies of a single random response does not add independent information. The value $1/20$ from part (a) would falsely count the copies as independent draws.

8. Calibrated measurements and estimation

(a) X is unbiased by assumption. Linearity and variance scaling give

$$\mathbb{E}[A] = \frac{(2\mu + 2) - 2}{2} = \mu, \qquad \mathrm{Var}(A) = \frac{1}{4}\mathrm{Var}(Y) = 1.$$

Both biases are zero, so their MSEs equal their variances: 4 for X and 1 for A. Thus A has smaller MSE under the stated model and squared-error criterion.

(b) Neither claim is justified by the information given. Unbiasedness establishes $\mathbb{E}[A] = \mu$ for the listed workers, not equality with the mean wage of all workers. If that broader target is μ_{all}, the bias for it is $\mu - \mu_{\text{all}}$; the question gives neither this difference nor its sign. The two means could happen to coincide. More independent measurements from the same list can improve precision for the list mean, but do not establish that the excluded workers have the same mean or remove the coverage gap. The estimator property and the population coverage problem must be assessed separately.

(c) The realized estimate is $a = (18 - 2)/2 = 8$. The true standard error of A is $\sqrt{\mathrm{Var}(A)} = 1$. The realized error is $8 - \mu$, which is unknown because μ is unknown. A standard error describes variability of the estimator across repetitions, not the signed or absolute error of this realized estimate.